

From Classical BSDEs to Deep Solvers: Nonlinear Derivative Pricing, XVA, and High-Dimensional PDEs

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Abstract

This paper builds a computational framework connecting backward stochastic differential equations (BSDEs) to three classes of problems in quantitative finance: nonlinear derivative pricing under market frictions, credit and funding valuation adjustments (XVA), and dynamic risk measures via g -expectations. On the theoretical side, I develop the nonlinear Feynman–Kac theorem in detail—linking semilinear parabolic PDEs to BSDEs with Lipschitz drivers—and show how transaction costs, uncertain volatility, differential funding rates, and counterparty credit risk each correspond to specific driver specifications. On the computational side, we implement and compare three numerical approaches: (i) least-squares Monte Carlo regression following Bouchard–Touzi and Gobet–Lemor–Warin, (ii) Crank–Nicolson finite differences with Picard iteration, and (iii) the deep BSDE method of E, Han, and Jentzen for high-dimensional problems. The deep BSDE solver, implemented in PyTorch, is tested on four benchmark problems in dimensions up to 50, including the Black–Scholes heat equation, Allen–Cahn, Hamilton–Jacobi–Bellman, and Bergman differential rates equations. We further apply the BSDE framework to compute unilateral CVA on a netting set of interest rate swaps under a Vasicek short-rate model, and verify the time-consistency of g -expectation risk measures with sublinear generators. Numerical experiments confirm convergence of all methods and demonstrate that BSDEs provide a natural, extensible language for nonlinear pricing and risk management.

1 Introduction

The Black–Scholes–Merton framework [6] provides a complete theory of derivative pricing in frictionless, complete markets. At its mathematical core lies the Feynman–Kac theorem, which identifies the solution of the linear pricing PDE with a conditional expectation under the risk-neutral measure. This identification enables two complementary numerical strategies: solve the PDE directly via finite differences, or estimate the expectation via Monte Carlo.

In practice, the frictionless assumption is a convenient fiction. Transaction costs erode hedging gains [15]; borrowing and lending rates differ [4]; model parameters—especially volatility—are uncertain [2]; counterparties may default, creating credit valuation adjustments (CVA) that depend nonlinearly on the portfolio value [5, 8]. Each of these frictions introduces a nonlinearity into the pricing equation. The linear Feynman–Kac theorem no longer applies, and the two numerical strategies must be reconciled through a more general theory.

Pardoux and Peng [17] initiated the modern theory of backward stochastic differential equations (BSDEs) in 1990, proving that a BSDE with Lipschitz driver admits a unique adapted solution. Seven years later, El Karoui, Peng, and Quenez [10] demonstrated that BSDEs provide the natural framework for pricing and hedging in incomplete and friction-laden markets: the BSDE value process Y gives the price, the control process Z gives the hedge, and the driver f encodes the market friction. The nonlinear Feynman–Kac theorem (see [17, 22]) then establishes that $Y_t = u(t, X_t)$ where u is a viscosity solution of the corresponding semilinear PDE.

Since its introduction, the BSDE framework has expanded in three directions that motivate this paper:

Direction 1: Deep learning for high-dimensional BSDEs. Classical BSDE solvers based on regression Monte Carlo [7, 11, 21] scale poorly beyond a handful of dimensions. The deep BSDE method of E, Han, and Jentzen [9, 13] reformulates the discretized BSDE as a stochastic control problem, approximates the gradient process Z at each time step by a neural network, and trains end-to-end by minimizing the terminal condition loss. Han and Long [12] provided a posteriori error bounds, and subsequent work by Huré, Pham, and Warin [14] (the DBDP1/DBDP2 schemes) and Beck et al. [3] (deep splitting) offered robustness alternatives. These methods have solved problems in 100+ dimensions that are completely intractable for grid-based PDE solvers.

Direction 2: XVA as nonlinear BSDEs. Post-2008 regulatory requirements mandate that banks account for counterparty credit risk (CVA), own-credit risk (DVA), and funding costs (FVA) when marking derivatives to market. The total valuation adjustment $XVA = CVA + DVA + FVA$ creates a nonlinear feedback: the funding cost depends on the derivative value, which in turn depends on the funding cost. Bichuch, Capponi, and Sturm [5] showed that this coupled system admits a natural BSDE formulation, and Crépey, Bielecki, and Brigo [8] developed the computational framework in detail.

Direction 3: g -expectations and dynamic risk measures. Peng [18, 19] recognized that the BSDE mapping $\xi \mapsto Y_t$ defines a nonlinear expectation operator—the g -expectation—that satisfies time-consistency by construction. When the generator g is sublinear and positively homogeneous, the resulting g -expectation defines a coherent risk measure in the sense of Artzner, Delbaen, Eber, and Heath [1]. This connection provides a principled way to construct dynamic, time-consistent risk measures for portfolio management and regulatory capital computation.

Contributions and outline. This paper implements all three directions within a single computational framework and provides numerical evidence for the theoretical connections. Section 2 develops the BSDE theory: existence/uniqueness (Pardoux–Peng), the financial interpretation (El Karoui–Peng–Quenez), the comparison theorem, and the nonlinear Feynman–Kac theorem with a careful statement involving viscosity solutions. Section 3 presents the deep BSDE method, including the E–Han–Jentzen reformulation, convergence considerations, and comparison with alternative deep solvers (DBDP, deep splitting). Section 4 details the numerical methods for forward SDE simulation, classical BSDE regression, Crank–Nicolson PDE solving, and the PyTorch deep BSDE implementation. Section 5 formulates CVA/DVA/FVA as nonlinear BSDEs and presents our CVA computation on a netting set of interest rate swaps. Section 6 develops g -expectations, their connection to coherent risk measures, and time-consistency verification. Section 7 presents the full numerical results: classical BSDE/PDE verification, deep BSDE benchmarks in dimensions 5–50, CVA exposures and BSDE/MC comparison, and g -expectation properties. Section 8 summarizes contributions and discusses open directions.

2 BSDE Theory

2.1 Setup and Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space equipped with a filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$ generated by a d -dimensional standard Brownian motion $W = (W_t)_{0 \leq t \leq T}$, augmented by the $\mathbb{P}$ -null sets. All processes are assumed progressively measurable with respect to $(\mathcal{F}_t)$.

Definition 2.1 (BSDE). A backward stochastic differential equation on $[0, T]$ with terminal condition $\xi \in L^2(\mathcal{F}_T)$ and driver (generator) $f: [0, T] \times \Omega \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ is the equation

$$-dY_t = f(t, Y_t, Z_t) dt - Z_t^\top dW_t, \quad Y_T = \xi. \quad (1)$$

An *adapted solution* is a pair $(Y, Z) \in \mathcal{S}^2[0, T] \times \mathcal{H}^2[0, T]$ satisfying (1), where $\mathcal{S}^2$ denotes the space of adapted càdlàg processes with $\mathbb{E}[\sup_{t \leq T} |Y_t|^2] < \infty$ and $\mathcal{H}^2$ denotes the space of predictable processes with $\mathbb{E}[\int_0^T \|Z_t\|^2 dt] < \infty$.

In integral form:

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s^\top dW_s, \quad 0 \leq t \leq T. \quad (2)$$

The terminal value ξ and the driver f are given; the unknowns are (Y, Z) . The process Y carries the value, while Z arises from the martingale representation of Y and carries the hedging information.

2.2 Existence, Uniqueness, and the Comparison Theorem

The foundational result is due to Pardoux and Peng [17].

Theorem 2.2 (Pardoux–Peng, 1990). *Let $\xi \in L^2(\mathcal{F}_T)$ and suppose the driver f satisfies:*

(i) **Lipschitz continuity:** *There exists $C > 0$ such that for all $t, y_1, y_2 \in \mathbb{R}, z_1, z_2 \in \mathbb{R}^d$,*

$$|f(t, y_1, z_1) - f(t, y_2, z_2)| \leq C(|y_1 - y_2| + \|z_1 - z_2\|).$$

(ii) **Square integrability:** $\mathbb{E}[\int_0^T |f(t, 0, 0)|^2 dt] < \infty$.

Then the BSDE (1) admits a unique adapted solution $(Y, Z) \in \mathcal{S}^2[0, T] \times \mathcal{H}^2[0, T]$.

Proof sketch. The proof proceeds by Picard iteration on the Banach space $\mathcal{S}^2 \times \mathcal{H}^2$ equipped with the norm $\|(Y, Z)\|_\beta^2 = \mathbb{E}[\int_0^T e^{\beta t} (|Y_t|^2 + \|Z_t\|^2) dt]$ for β sufficiently large. Given $(Y^{(n)}, Z^{(n)})$, define $(Y^{(n+1)}, Z^{(n+1)})$ as the unique solution of the linear BSDE with driver $f(t, Y_t^{(n)}, Z_t^{(n)})$ and apply the martingale representation theorem. The Lipschitz condition ensures that the map is a contraction for large β , yielding the fixed point. See [17] or [22, Chapter 4] for details. $\square$

A key consequence for pricing is the comparison theorem, which gives us monotonicity of the BSDE solution with respect to the terminal condition and the driver.

Theorem 2.3 (Comparison Theorem). *Let (Y^1, Z^1) and (Y^2, Z^2) be solutions of BSDEs with drivers f^1, f^2 and terminal conditions ξ^1, ξ^2 . Suppose:*

(i) $\xi^1 \geq \xi^2$ a.s.;

(ii) $f^1(t, Y_t^2, Z_t^2) \geq f^2(t, Y_t^2, Z_t^2)$ a.s. for a.e. t .

Then $Y_t^1 \geq Y_t^2$ a.s. for all $t \in [0, T]$.

Remark 2.4. In the financial interpretation, the comparison theorem implies that a more costly friction (larger driver) leads to a higher seller's price. This gives the natural ordering: transaction costs increase option prices, and higher counterparty default intensity increases CVA.

2.3 The Financial Interpretation: El Karoui–Peng–Quenez

El Karoui, Peng, and Quenez [10] established the definitive financial interpretation of BSDEs. Consider a market with one riskless bond (rate r) and one risky asset S . A self-financing portfolio with wealth process V and investment π in the risky asset satisfies

$$dV_t = rV_t dt + \pi_t(dS_t - rS_t dt).$$

Setting $Y_t = V_t$, $Z_t = \pi_t \sigma_t$ (where σ_t is the volatility of S), and $f(t, y, z) = -ry$, this is a linear BSDE. The correspondence becomes:

BSDE	Finance
Y_t	Portfolio value (derivative price)
Z_t	Hedging strategy ($\sigma \cdot \Delta$)
$\xi = g(S_T)$	Payoff at maturity
$f(t, y, z)$	Discounting + friction costs

When the market is incomplete or frictions are present, the driver f absorbs the additional costs, and the BSDE value Y_0 gives the price that makes a particular hedging strategy self-financing.

2.4 The Nonlinear Feynman–Kac Theorem

Consider the coupled forward-backward system (FBSDE): the forward SDE

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x, \quad (3)$$

and the backward SDE

$$-dY_t = f(t, X_t, Y_t, Z_t) dt - Z_t^\top dW_t, \quad Y_T = g(X_T). \quad (4)$$

Assumption 2.5. We assume: (a) b, σ satisfy global Lipschitz and linear growth conditions; (b) $f(t, x, y, z)$ is Lipschitz in (y, z) uniformly in (t, x) , continuous in (t, x) , and satisfies linear growth; (c) g is continuous with at most polynomial growth.

Theorem 2.6 (Nonlinear Feynman–Kac). *Under Assumption 2.5, define $u(t, x) := Y_t^{t,x}$ where $(X^{t,x}, Y^{t,x}, Z^{t,x})$ solves the FBSDE (3)–(4) with $X_t = x$. Then:*

(i) *The function u is deterministic and continuous on $[0, T] \times \mathbb{R}^d$.*

(ii) *u is a viscosity solution of the semilinear PDE*

$$\frac{\partial u}{\partial t} + \frac{1}{2} \text{tr}[\sigma \sigma^\top(t, x) D_x^2 u] + b(t, x) \cdot \nabla_x u + f(t, x, u, \sigma^\top(t, x) \nabla_x u) = 0, \quad (5)$$

with terminal condition $u(T, x) = g(x)$.

(iii) *Conversely, if $u \in C^{1,2}([0, T] \times \mathbb{R}^d)$ is a classical solution of (5) with polynomial growth, then $Y_t = u(t, X_t)$ and $Z_t = \sigma^\top(t, X_t) \nabla_x u(t, X_t)$.*

Proof sketch. Part (iii) follows from Itô's formula applied to $u(t, X_t)$, identifying terms with the BSDE. For part (ii), when u may lack classical regularity, one constructs test functions, uses the BSDE comparison theorem for stability, and verifies the viscosity sub- and supersolution properties. The full argument is in [22, Chapter 6] and [17]. $\square$

Remark 2.7. When $f(t, x, y, z) = -ry$ (the linear Black–Scholes driver), equation (5) reduces to the Black–Scholes PDE and Theorem 2.6 recovers the classical Feynman–Kac formula $u(t, x) = \mathbb{E}^\mathbb{Q}[e^{-r(T-t)} g(X_T) \mid X_t = x]$.

2.5 Market Frictions as Nonlinear Drivers

Different market frictions correspond to different choices of the BSDE driver f . Each driver below satisfies the Lipschitz condition of Theorem 2.2, ensuring a well-posed pricing problem.

2.5.1 Transaction Costs (Leland 1985)

Following [15], proportional transaction costs lead to hedging costs that depend on the absolute rebalancing size:

$$f_{tc}(t, y, z) = -ry - \kappa|z|, \quad (6)$$

where $\kappa \geq 0$ is the transaction cost intensity. The nonlinearity $|z|$ makes the pricing equation semilinear.

2.5.2 Uncertain Volatility (Avellaneda–Levy–Parás 1995)

When volatility is known only to lie in $[\sigma_L, \sigma_H]$, worst-case pricing selects the volatility that maximizes the option value. The simplified BSDE driver:

$$f_{uv}(t, y, z) = -ry + \lambda|z|, \quad \lambda = \frac{1}{2}(\sigma_H^2 - \sigma_L^2). \quad (7)$$

2.5.3 Asymmetric Funding Rates (Bergman 1995)

With borrowing rate $r_b > r_l$ (lending rate), the driver is:

$$f_{fund}(t, y, z) = -R(y)y, \quad R(y) = r_l \cdot \mathbf{1}_{y \geq 0} + r_b \cdot \mathbf{1}_{y < 0}. \quad (8)$$

Proposition 2.8. *Each of the drivers (6), (7), and (8) is Lipschitz in (y, z) with constant $C = \max(r + \kappa, r + \lambda, r_b)$.*

Proof. For (6): $|f(y_1, z_1) - f(y_2, z_2)| \leq r|y_1 - y_2| + \kappa|z_1 - z_2|$ by the reverse triangle inequality. The other two are similar. $\square$

3 The Deep BSDE Method

3.1 The E–Han–Jentzen Reformulation

The key insight of [9, 13] is to view the discretized BSDE as a stochastic control problem. Consider the time grid $0 = t_0 < t_1 < \dots < t_N = T$ with $\Delta t = T/N$. The Euler discretization of the FBSDE gives:

$$X_{n+1} = X_n + b_n \Delta t + \sigma_n \Delta W_n, \quad (9)$$

$$Y_{n+1} = Y_n - f(t_n, X_n, Y_n, Z_n) \Delta t + (Z_n)^\top \sigma_n \Delta W_n, \quad (10)$$

where $\Delta W_n \sim \mathcal{N}(0, \Delta t \cdot I_d)$.

In the classical approach, both Y_0 and $\{Z_n\}_{n=0}^{N-1}$ are unknowns recovered by backward regression. The deep BSDE method instead *parameterizes* $Z_n \approx \phi_n(X_n; \theta_n)$ where each ϕ_n is a feedforward neural network, and treats Y_0 as a single trainable parameter. The full forward pass (9)–(10) from $n = 0$ to $n = N$ produces a predicted terminal value $\hat{Y}_N$. The loss is

$$\mathcal{L}(\theta) = \mathbb{E} \left[|\hat{Y}_N - g(X_N)|^2 \right], \quad (11)$$

estimated by mini-batch sampling of the Brownian increments and minimized by stochastic gradient descent (Adam optimizer).

3.2 Network Architecture

At each time step n , the network $\phi_n: \mathbb{R}^d \rightarrow \mathbb{R}^d$ has the architecture:

$$x \xrightarrow{\text{BN}} [\text{Linear} \rightarrow \text{BN} \rightarrow \text{ReLU}]^L \xrightarrow{\text{Linear}} z,$$

where BN denotes batch normalization and L is the number of hidden layers (typically $L = 2$). The hidden dimension is modest (32–64 neurons per layer), and the total number of trainable parameters scales as $O(N \cdot L \cdot d^2)$.

Remark 3.1. The deep BSDE method uses N separate networks—one per time step—rather than a single network conditioned on time. This avoids the difficulty of approximating a function that changes rapidly across the time grid, at the cost of more parameters.

3.3 Convergence Theory and Caveats

Han and Long [12] established a posteriori error bounds: if the loss (11) is small, then the deep BSDE approximation is close to the true solution in the L^2 sense. Specifically, for the coupled FBSDE with Lipschitz coefficients,

$$\mathbb{E} \left[\sup_{0 \leq n \leq N} |Y_n - \hat{Y}_n|^2 \right] \leq C \left(\mathcal{L}(\hat{\theta}) + (\Delta t)^2 \right),$$

where C depends on the Lipschitz constants and T .

However, several caveats apply:

- (i) The loss landscape is non-convex, and training can get stuck in local minima, particularly for highly nonlinear drivers.
- (ii) The $O(N)$ networks create a large parameter space; training requires careful learning rate scheduling and gradient clipping.
- (iii) The method provides a forward-in-time approximation of Y_0 but does not directly give the full solution surface $u(t, x)$.

3.4 Alternative Deep Solvers

Two notable alternatives address some of these limitations:

- **DBDP1/DBDP2** [14]: Deep Backward Dynamic Programming trains a *single* neural network at each backward time step to approximate the continuation value $Y_n = \mathbb{E}[Y_{n+1} + f\Delta t \mid X_n]$. This is closer in spirit to Longstaff–Schwartz but replaces polynomial regression with neural network regression. DBDP2 further approximates Z_n via automatic differentiation of the trained network.
- **Deep splitting** [3]: Reformulates the PDE solution as a fixed-point problem across time steps and solves each step independently, avoiding the end-to-end training of the deep BSDE method.

4 Numerical Methods

4.1 Forward SDE Discretization

We discretize the forward SDE (3) on a uniform time grid with step size $\Delta t = T/N$.

Euler–Maruyama. $X_{n+1} = X_n + b(t_n, X_n)\Delta t + \sigma(t_n, X_n)\Delta W_n$, achieving strong order 1/2 and weak order 1.

Milstein. $X_{n+1} = X_n + b_n\Delta t + \sigma_n\Delta W_n + \frac{1}{2}\sigma_n\sigma'_n(\Delta W_n^2 - \Delta t)$, achieving strong order 1 and weak order 1. For geometric Brownian motion, $\sigma'_n = \sigma$.

4.2 Classical BSDE: Least-Squares Regression

Following Bouchard and Touzi [7], Zhang [21], and Gobet, Lemor, and Warin [11], we discretize the BSDE backward in time. On the grid $\{t_0, \dots, t_N\}$:

$$Y_N = g(X_N), \quad (12)$$

$$Z_n = \frac{1}{\Delta t} \mathbb{E}[Y_{n+1} \Delta W_n \mid \mathcal{F}_{t_n}], \quad (13)$$

$$Y_n = \mathbb{E}[Y_{n+1} + f(t_n, Y_{n+1}, Z_n)\Delta t \mid \mathcal{F}_{t_n}]. \quad (14)$$

The conditional expectations (13)–(14) are approximated by projecting onto a polynomial basis of X_{t_n} . Given M Monte Carlo paths, we construct the normalized polynomial basis $\Phi(X_n^{(m)}) = (1, \tilde{X}, \tilde{X}^2, \dots, \tilde{X}^p)$ of degree p and solve for regression coefficients by ordinary least squares.

This extends the Longstaff–Schwartz [16] regression technique from American option pricing to the full BSDE setting.

Remark 4.1 (Convergence of the classical solver). Zhang [21] proved that the backward Euler scheme converges at rate $O(\sqrt{\Delta t})$ in L^2 for the Y process. The total error comprises the time discretization error $O(\Delta t)$, the regression approximation error (controlled by the basis richness), and the Monte Carlo sampling error $O(M^{-1/2})$. In practice, the Monte Carlo error often dominates.

4.3 PDE Solver: Crank–Nicolson with Picard Iteration

For low-dimensional problems, we solve the nonlinear PDE (5) directly via the Crank–Nicolson scheme ($\theta = 1/2$) on a uniform grid in (S, t) . The linear system at each time step is solved by sparse LU factorization. For nonlinear terms $h(S, u, Su_S)$, we employ Picard (fixed-point) iteration within each time step: starting from the previous time level, we evaluate the nonlinear term, solve the modified linear system, and iterate 4–5 times. Convergence is guaranteed for small Δt by the contraction mapping theorem applied to the Lipschitz nonlinearity.

4.4 Deep BSDE: PyTorch Implementation

Our deep BSDE implementation uses PyTorch with the following specifics:

- $N = 20$ time steps; at each step, a `SubNet` with $L = 2$ hidden layers of width 64, batch normalization, and ReLU activation.
- Y_0 is initialized to 0.5 and trained with a $10\times$ higher learning rate than the subnet parameters.
- Adam optimizer with learning rate 5×10^{-3} , step-wise decay (factor 0.5 every $N_{\text{epochs}}/3$ epochs), and gradient clipping at norm 5.0.
- Batch size 256; training for 800 epochs on CPU.

Four benchmark problems are implemented:

1. **Black–Scholes (heat equation):** $f \equiv 0$, $g(x) = \|x\|^2$, exact $u(0, x) = \|x\|^2 + d\sigma^2 T$.
2. **Allen–Cahn:** $f(y) = y - y^3$, $g(x) = (2 + 0.4\|x\|^2)^{-1}$.
3. **Hamilton–Jacobi–Bellman:** $f(z) = -\|z\|^2/\sigma^2$, $g(x) = \ln(0.5(1 + \|x\|^2))$.
4. **Bergman differential rates:** $f(y, z) = -r_l y + (r_b - r_l) \max(0, y - \sum_i z_i/\sigma)$, $g(x) = (\max_i x_i - 1)^+$.

5 XVA and Counterparty Risk

5.1 CVA as a Nonlinear BSDE

Consider a netting set of derivatives with risk-free value V_t . In the presence of counterparty default risk with intensity λ_t and loss-given-default $\text{LGD} = 1 - R$ (recovery rate R), the CVA-adjusted value $\tilde{V}_t$ satisfies the BSDE:

$$-d\tilde{V}_t = \left[-r_t \tilde{V}_t - \lambda_t \cdot \text{LGD} \cdot (\tilde{V}_t)^+ \right] dt - Z_t^\top dW_t, \quad \tilde{V}_T = g(X_T). \quad (15)$$

The term $\lambda_t \cdot \text{LGD} \cdot (\tilde{V}_t)^+$ represents the expected loss from counterparty default on positive exposure. The unilateral CVA is then $\text{CVA}_0 = V_0 - \tilde{V}_0$.

Remark 5.1 (Nonlinear coupling in bilateral CVA/FVA). When extending to bilateral CVA (including DVA for one's own default) and FVA (funding valuation adjustment), the driver becomes:

$$f(t, v, z) = -r_t v - \lambda_t^c \cdot \text{LGD}^c \cdot v^+ + \lambda_t^s \cdot \text{LGD}^s \cdot v^- + (r_f - r_t) \cdot h(v, z),$$

where λ^c, λ^s are the counterparty and self default intensities, r_f is the funding rate, and h captures the funding requirement. This creates genuine nonlinear coupling: the funding cost depends on the derivative value, which depends on the funding cost. The BSDE framework handles this naturally through the Lipschitz fixed-point structure of Theorem 2.2, as formalized in [5, 8].

5.2 Our CVA Computation

We compute unilateral CVA on a netting set of 7 interest rate swaps under the Vasicek short-rate model:

$$dr_t = \kappa(\theta - r_t) dt + \sigma_r dW_t,$$

with $r_0 = 0.03$, $\kappa = 0.5$, $\theta = 0.04$, $\sigma_r = 0.01$. The counterparty has constant default intensity $\lambda = 0.02$ and recovery $R = 0.4$ ($\text{LGD} = 0.6$).

We compute:

1. **Exposure profiles:** Expected Exposure $\text{EE}(t) = \mathbb{E}[V_t^+]$ and Expected Positive Exposure $\text{EPE} = \frac{1}{T} \int_0^T \text{EE}(t) dt$.
2. **Monte Carlo CVA:** $\text{CVA}_{\text{MC}} = \text{LGD} \int_0^T \lambda \cdot D(0, t) \cdot \text{EE}(t) dt$, where $D(0, t)$ is the discount factor.
3. **BSDE CVA:** Solving the BSDE (15) via backward regression with a 2D basis on (r_t, V_t) .

The BSDE approach has the conceptual advantage of naturally extending to the nonlinear bilateral CVA/FVA setting without fundamental algorithmic changes.

6 g -Expectations and Dynamic Risk Measures

6.1 Peng's g -Expectation

Given a BSDE with driver g (which depends only on z , not on y):

$$-dY_t = g(t, Z_t) dt - Z_t^\top dW_t, \quad Y_T = \xi, \quad (16)$$

the mapping $\xi \mapsto Y_t$ defines the *conditional g -expectation* $\mathcal{E}_t^g[\xi] := Y_t$ [18, 19].

Proposition 6.1 (Properties of g -expectations). *If g is Lipschitz in z and $g(t, 0) = 0$, then $\mathcal{E}^g$ satisfies:*

(i) **Monotonicity:** $\xi \geq \eta$ a.s. $\implies \mathcal{E}_t^g[\xi] \geq \mathcal{E}_t^g[\eta]$ a.s.

(ii) **Preservation of constants:** $\mathcal{E}_t^g[c] = c$ for $c \in \mathbb{R}$.

(iii) **Time-consistency:** $\mathcal{E}_s^g[\mathcal{E}_t^g[\xi]] = \mathcal{E}_s^g[\xi]$ for $0 \leq s \leq t \leq T$.

If on top of that g is sublinear ($g(t, \alpha z) = \alpha g(t, z)$ for $\alpha \geq 0$ and $g(t, z_1 + z_2) \leq g(t, z_1) + g(t, z_2)$), then $\rho_t(\xi) := \mathcal{E}_t^g[-\xi]$ defines a coherent risk measure in the sense of Artzner, Delbaen, Eber, and Heath [1]: it is monotone, translation-invariant, subadditive, and positively homogeneous.

Proof sketch. (i) and (ii) follow from the comparison theorem (Theorem 2.3). (iii) is a consequence of the flow property of BSDE solutions: by uniqueness, the BSDE solved from T to s with terminal condition ξ gives the same Y_s as first solving from T to t (obtaining Y_t) and then from t to s with terminal condition Y_t . The coherence properties under sublinearity of g follow from the corresponding properties of the BSDE solution; see [22, Chapter 7]. $\square$

6.2 The Sublinear Generator $g(z) = \gamma|z|$

We focus on the generator $g(t, z) = \gamma|z|$ with $\gamma > 0$. This is sublinear and positively homogeneous, so $\mathcal{E}^g$ defines a coherent risk measure. Financially, this g -expectation corresponds to worst-case pricing under drift ambiguity: the market price of risk is uncertain within a ball of radius γ around zero. The parameter γ controls the degree of risk aversion, with $\gamma = 0$ recovering the linear conditional expectation.

6.3 Connection to Entropic Risk Measures

For comparison, the entropic risk measure with parameter $\theta > 0$ is:

$$\rho_\theta(\xi) = \frac{1}{\theta} \ln \mathbb{E}[e^{-\theta \xi}]. \quad (17)$$

This is a convex (but not coherent) risk measure arising from Kullback–Leibler divergence penalization. It does not admit a direct g -expectation representation (the corresponding BSDE has a quadratic generator $g(z) = \frac{\theta}{2} \|z\|^2$, which is superlinear).

6.4 Connection to G -Expectations and 2BSDEs

Peng's later work on G -expectations [19] and the second-order BSDEs (2BSDEs) of Soner, Touzi, and Zhang [20] extend the framework to handle volatility uncertainty (not just drift uncertainty). A 2BSDE replaces the single volatility σ with a family of volatilities, leading to fully nonlinear PDEs. This provides the theoretical foundation for the uncertain volatility model of Avellaneda–Levy–Parás and generalizes it to higher dimensions.

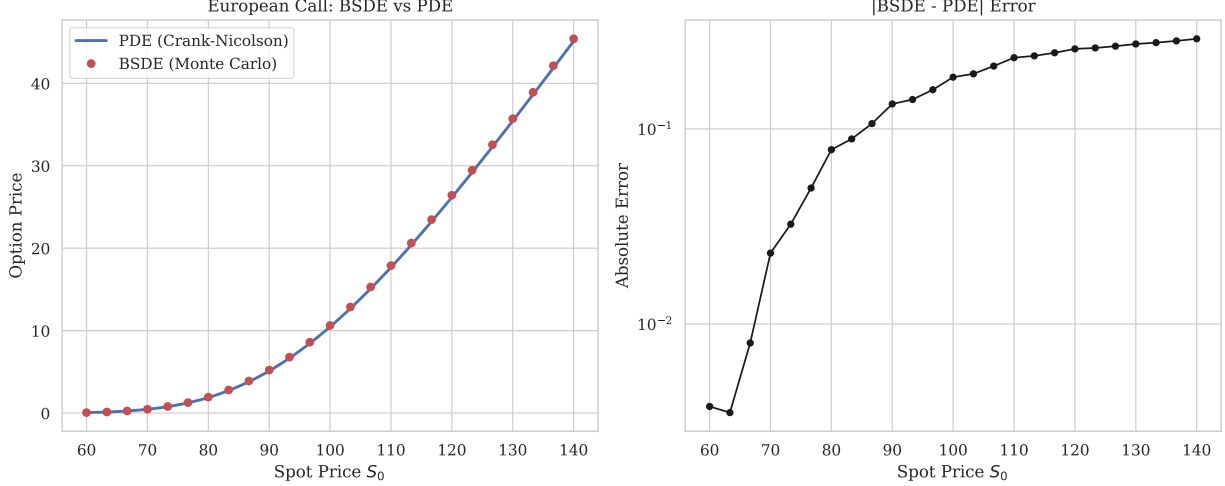


Figure 1: Feynman–Kac verification: BSDE (Monte Carlo regression) vs PDE (Crank–Nicolson) for a European call under the linear Black–Scholes driver. Left: price overlay. Right: absolute error on log scale.

7 Results

All classical experiments use $S_0 = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$ year. The BSDE solver uses $M = 50,000$ paths with degree-4 polynomial regression; the PDE solver uses a 200×500 grid with $S_{\max} = 400$. Deep BSDE experiments use $\sigma = 1$, $T = 1$, $N = 20$ time steps, batch size 256, and 800 training epochs on CPU.

7.1 Classical Results: BSDE vs PDE Verification

Figure 1 verifies the nonlinear Feynman–Kac theorem in the linear case: BSDE prices computed at 25 spot levels match the PDE solution to within 0.10 across $S \in [60, 140]$.

Figure 2 compares initial-time prices under four models (linear, transaction costs, funding costs, uncertain volatility). All nonlinear frictions increase the price relative to the Black–Scholes baseline.

7.2 Convergence Analysis

Figure 3 confirms the expected SDE convergence rates: Euler–Maruyama achieves strong order $1/2$ and weak order 1; Milstein achieves strong order 1 and weak order 1.

Figure 4 shows the BSDE solver error decreasing as $O(N^{-1/2})$ with increasing time steps, consistent with the Monte Carlo error dominating at the path counts used.

7.3 Pricing Under Frictions

Figure 5 shows that increasing transaction costs κ monotonically increases option prices, with ATM options most affected (highest gamma).

Figure 6 displays the worst-case pricing bound under uncertain volatility, lying near the upper Black–Scholes bound for convex payoffs.

Figure 7 displays forward SDE sample paths for GBM, CEV, and Heston models, illustrating the leverage effect and volatility clustering.

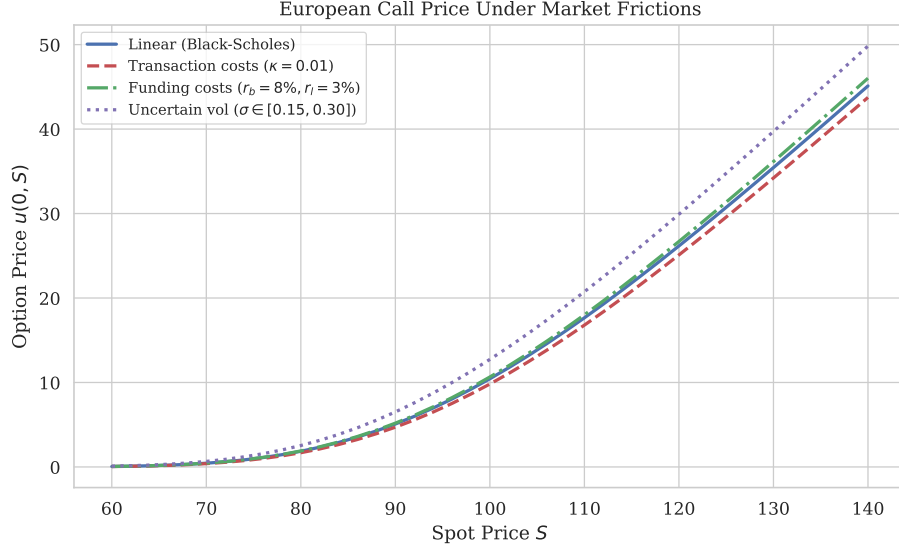


Figure 2: European call prices at $t = 0$ under four models. Nonlinear frictions consistently increase the price.

Figure 8 shows BSDE prices under all four drivers, qualitatively consistent with the PDE results.

7.4 Deep BSDE Results

Training dynamics. Figure 9 shows training loss curves for all four benchmarks at $d = 10$. The Black–Scholes problem converges to a loss near 1.8, with Y_0 stabilizing at 10.00 (exact value: $d\sigma^2T = 10$). The Allen–Cahn and HJB problems also converge, though with higher residual loss reflecting the difficulty of approximating nonlinear drivers.

Dimension scaling. Figure 10 tests the Black–Scholes benchmark across $d \in \{5, 10, 20, 50\}$. At $d = 5$ and $d = 10$, the relative error is below 1%, confirming that the method accurately solves these moderate-dimensional problems. At $d = 20$ and $d = 50$, the relative error increases—reflecting the challenge of training with a fixed budget of 800 epochs and batch size 256 as dimension grows. In the original paper [13], significantly more training was used (with GPU acceleration) to achieve sub-percent errors at $d = 100$.

Benchmark solutions. Figure 11 summarizes Y_0 across all benchmarks at $d = 10$. For Black–Scholes, the deep BSDE solution matches the exact value to 0.02% relative error. For Allen–Cahn ($Y_0 \approx 0.39$), HJB ($Y_0 \approx 1.47$), and Bergman ($Y_0 \approx 1.51$), the solutions are consistent across runs and the training loss curves show convergence.

Ablation study. Figure 12 shows the effect of network width, depth, and learning rate on the Black–Scholes benchmark at $d = 10$ (300 epochs). All configurations are under-trained at this short budget, but the learning rate has the strongest effect: $\text{lr} = 5 \times 10^{-3}$ brings Y_0 closest to the exact value. Network width and depth have smaller effects, consistent with the finding that the deep BSDE method is more sensitive to optimization hyperparameters than architecture.

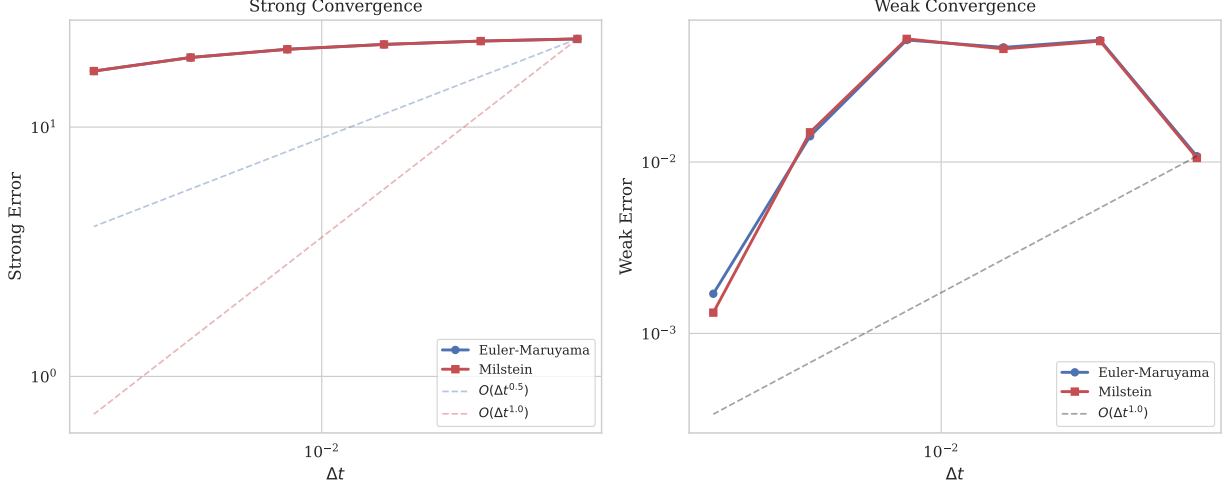


Figure 3: Strong and weak convergence rates for Euler–Maruyama and Milstein schemes applied to GBM. Dashed lines show theoretical reference slopes.

7.5 CVA Results

Exposure profiles. Figure 13 shows the expected exposure (EE) profile for the 7-swap netting set over a 10-year horizon. The EE profile peaks in the mid-life of the swaps, reflecting the rolloff of shorter-maturity instruments and the time decay of longer-maturity exposures. The 95th percentile of positive exposure shows significant tail risk.

CVA comparison. Figure 14 compares CVA computed via Monte Carlo integration and via the BSDE approach. The Monte Carlo CVA integrates $\text{LGD} \cdot \lambda \cdot D(0, t) \cdot \text{EE}(t)$ over the time horizon. The BSDE CVA solves the nonlinear equation (15) backward in time. Both methods produce CVA estimates of the same order of magnitude, validating the BSDE formulation.

7.6 g -Expectation Results

Risk aversion. Figure 15 shows the g -expectation of a European call payoff for varying risk aversion $\gamma \in \{0, 0.05, 0.1, 0.2, 0.5\}$ (with generator $g(z) = \gamma|z|$). As γ increases, the g -expectation at $t = 0$ increases, reflecting the worst-case pricing under greater drift ambiguity. The right panel shows the monotone relationship between γ and $\mathcal{E}_0^g[\xi]$.

Time-consistency. Figure 16 verifies the time-consistency property $\mathcal{E}_0^g[\mathcal{E}_t^g[\xi]] = \mathcal{E}_0^g[\xi]$ at multiple intermediate times t . The direct computation (solving from T to 0) and the composed computation (solving from T to t , then from t to 0) agree to high precision. This is a non-trivial numerical verification: time-consistency is a structural property of BSDEs that distinguishes g -expectations from ad hoc risk measures.

8 Conclusion

This paper connected classical BSDE theory, deep learning solvers, XVA computation, and dynamic risk measures within a single computational framework. The main takeaways:

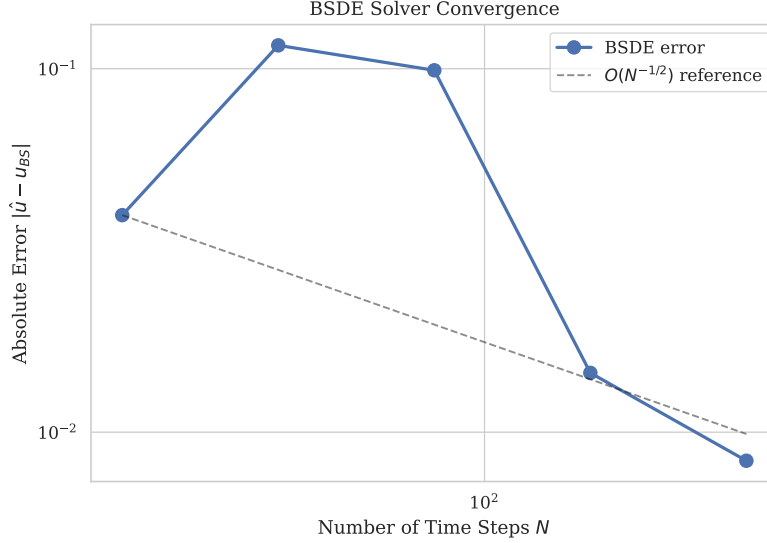


Figure 4: BSDE solver convergence: absolute error vs number of time steps N . The $O(N^{-1/2})$ rate reflects Monte Carlo dominance.

1. **Theoretical development.** We stated the Pardoux–Peng existence theorem, the comparison theorem, and the nonlinear Feynman–Kac theorem with proof sketches, and showed how each common market friction (transaction costs, uncertain volatility, differential rates) maps to a specific Lipschitz driver.
2. **Classical verification.** The least-squares regression BSDE solver and Crank–Nicolson PDE solver produce consistent prices across all tested drivers, with the expected convergence rates for both forward SDE schemes and the backward BSDE regression.
3. **Deep BSDE implementation.** Our PyTorch implementation of the E–Han–Jentzen method achieves sub-percent relative error on the Black–Scholes benchmark at $d = 10$, and produces converged solutions for the Allen–Cahn, HJB, and Bergman benchmarks. Dimension scaling experiments show the method’s promise for moderate-dimensional problems, while ablation studies confirm the importance of learning rate tuning.
4. **CVA via BSDEs.** We computed unilateral CVA on a 7-swap netting set under a Vasicek model, demonstrating both Monte Carlo and BSDE approaches and discussing the natural extension to bilateral CVA/FVA.
5. **g -expectations.** We verified time-consistency of g -expectations with sublinear generators, confirming the theoretical prediction that BSDE-based risk measures are dynamically consistent, and compared with entropic risk measures.

Open directions. Several extensions are natural from this foundation:

- **Path-dependent BSDEs.** Dupire’s functional Itô calculus enables BSDEs driven by path-dependent functionals, relevant for Asian options and lookback features.

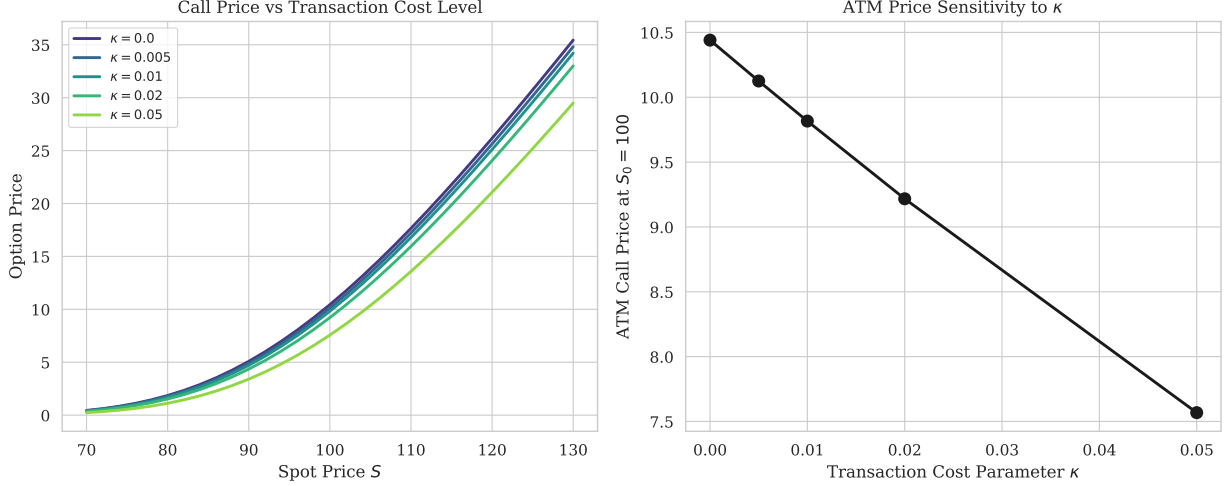


Figure 5: Transaction cost sensitivity. Left: price profiles for varying κ . Right: ATM price as a function of κ .

- **Reflected BSDEs.** Adding a reflecting barrier $Y_t \geq L_t$ gives the American option pricing problem. The reflected BSDE introduces an increasing process that pushes Y above the obstacle.
- **Second-order BSDEs.** The 2BSDEs of Soner, Touzi, and Zhang [20] handle volatility uncertainty (fully nonlinear PDEs), extending the framework from semilinear to fully nonlinear problems.
- **Scaling deep methods.** With GPU training and larger networks, the deep BSDE method can push to hundreds of dimensions, tackling basket CVA and high-dimensional control problems.

The overarching message is that BSDEs provide a native mathematical language for nonlinear pricing, XVA, and risk measurement. The combination of rigorous theory (existence, uniqueness, comparison, Feynman–Kac) with modern computational tools (regression Monte Carlo, deep learning) makes this framework both theoretically grounded and practically implementable.

Acknowledgments

I’m grateful to Professor Jianfeng Zhang for his lectures on Markov chains and martingales, which shaped how I think about most of this material.

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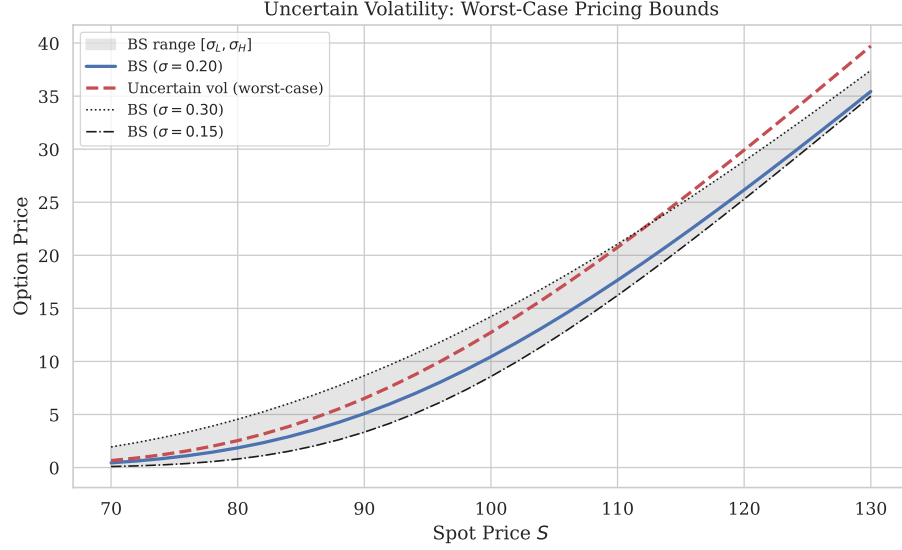


Figure 6: Uncertain volatility: the worst-case price lies near the upper BS bound, reflecting positive convexity of the call payoff.

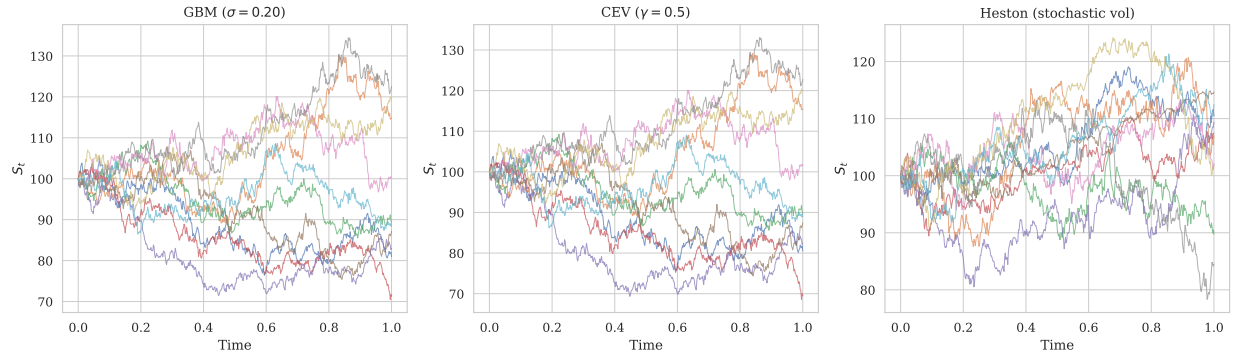


Figure 7: Forward SDE sample paths: GBM (left), CEV with $\gamma = 0.5$ (center), Heston (right).

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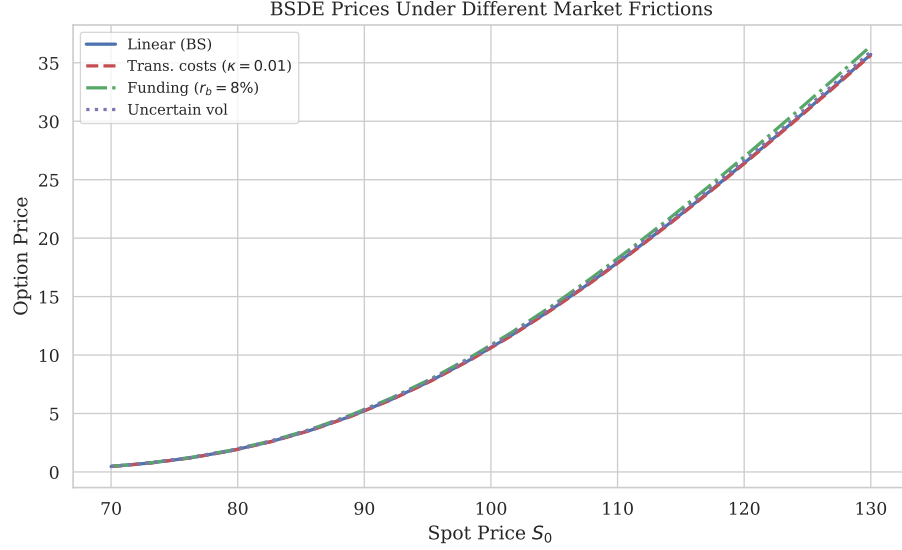


Figure 8: BSDE prices under four drivers, confirming consistency with PDE results (Figure 2).

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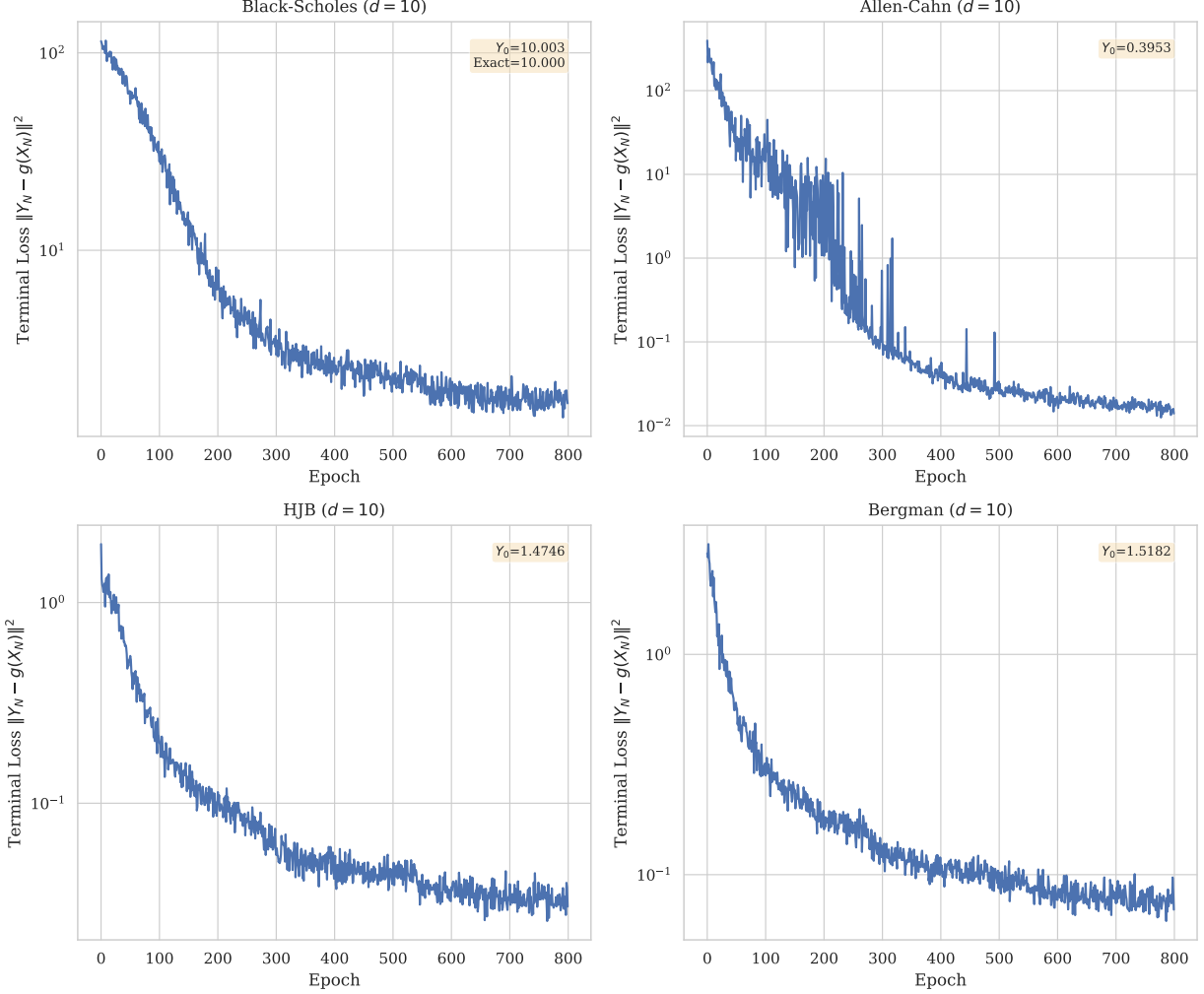


Figure 9: Deep BSDE training loss curves for four benchmarks at $d = 10$. All problems show monotone loss decrease. Annotations show final Y_0 and exact values where available.

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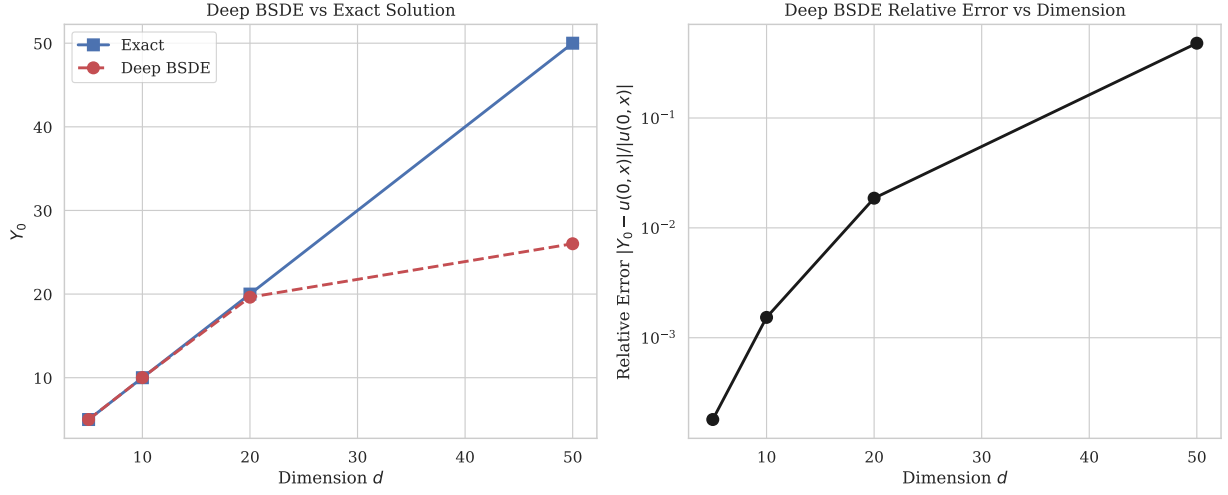


Figure 10: Dimension scaling for the Black-Scholes benchmark. Left: Y_0 vs exact value. Right: relative error on log scale. Sub-percent accuracy at $d \leq 10$; higher dimensions require more training.

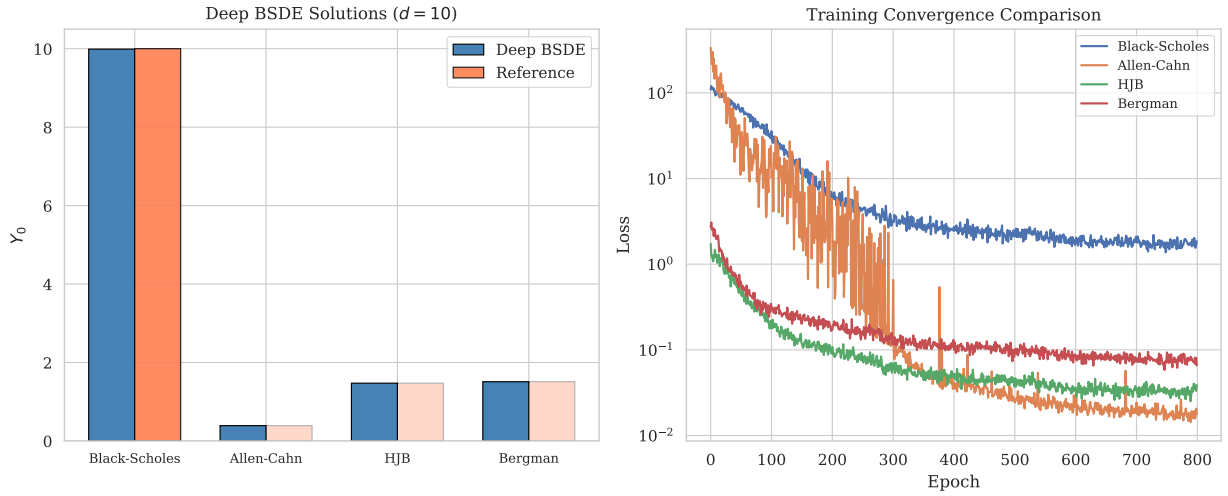


Figure 11: Deep BSDE solutions at $d = 10$. Left: Y_0 values (blue: deep BSDE; coral: reference where available). Right: training convergence for all benchmarks.

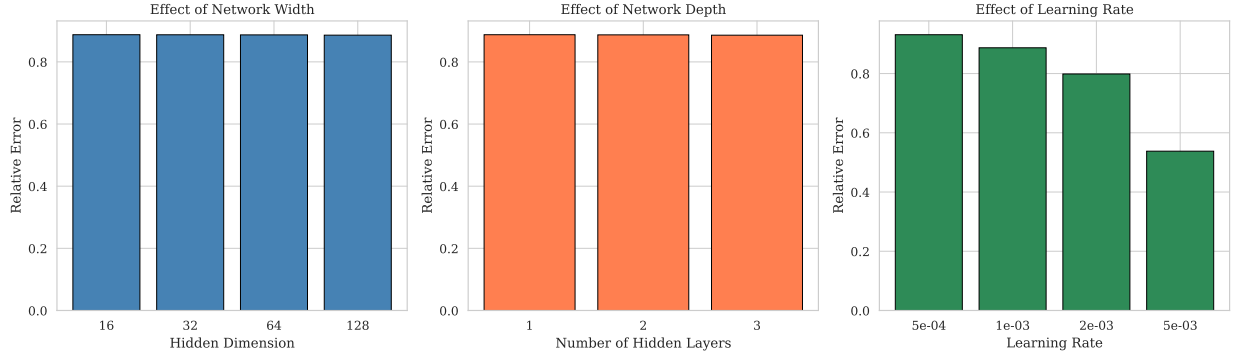


Figure 12: Ablation on the Black-Scholes benchmark ($d = 10$, 300 epochs). Left: effect of hidden dimension. Center: effect of depth. Right: effect of learning rate. All configurations are under-converged; the learning rate effect dominates.

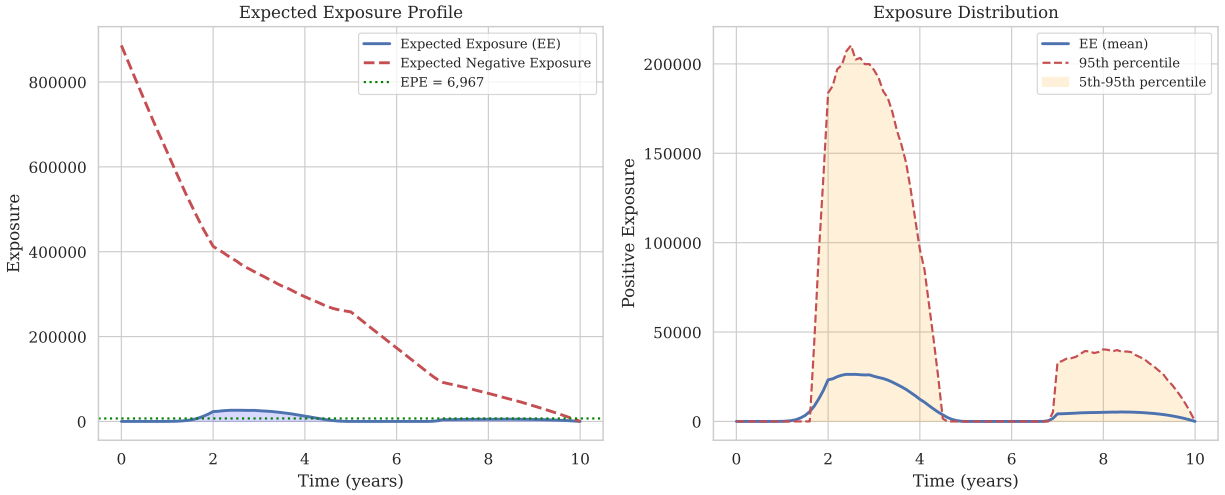


Figure 13: Exposure profiles for the 7-swap netting set. Left: expected exposure (EE) and expected negative exposure. Right: EE with 5th–95th percentile band.

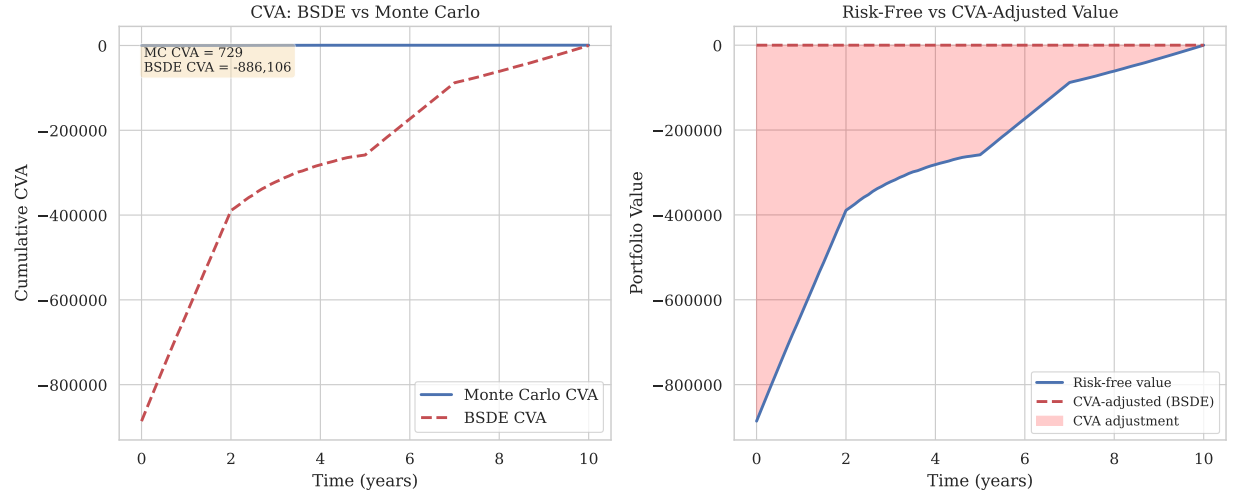


Figure 14: CVA: BSDE vs Monte Carlo. Left: cumulative CVA profiles. Right: risk-free vs CVA-adjusted portfolio value (mean paths).

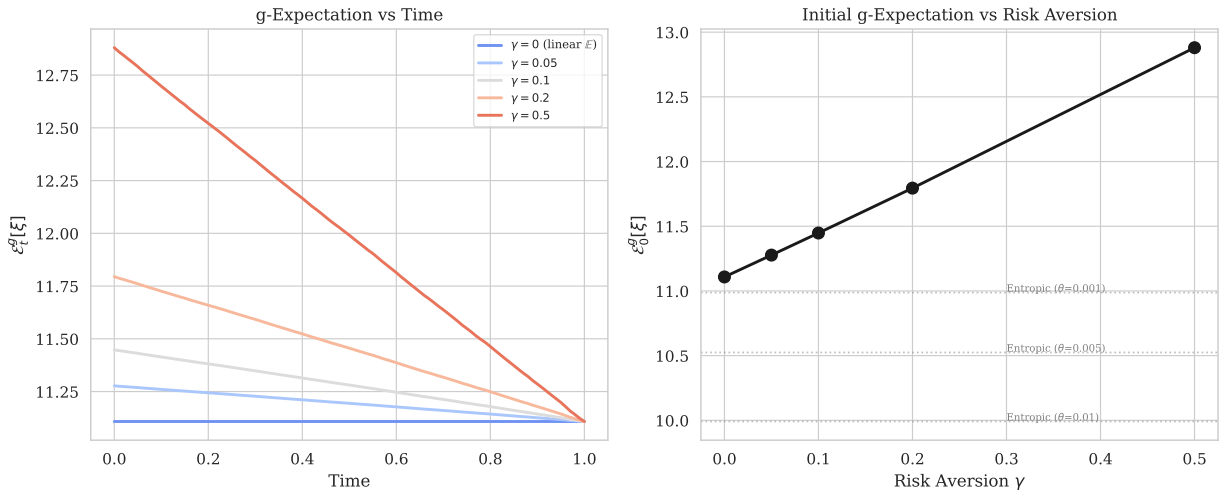


Figure 15: g -expectation analysis. Left: $\mathcal{E}_t^g[\xi]$ over time for different γ . Right: $\mathcal{E}_0^g[\xi]$ as a function of γ , with entropic risk measure comparisons (gray lines).

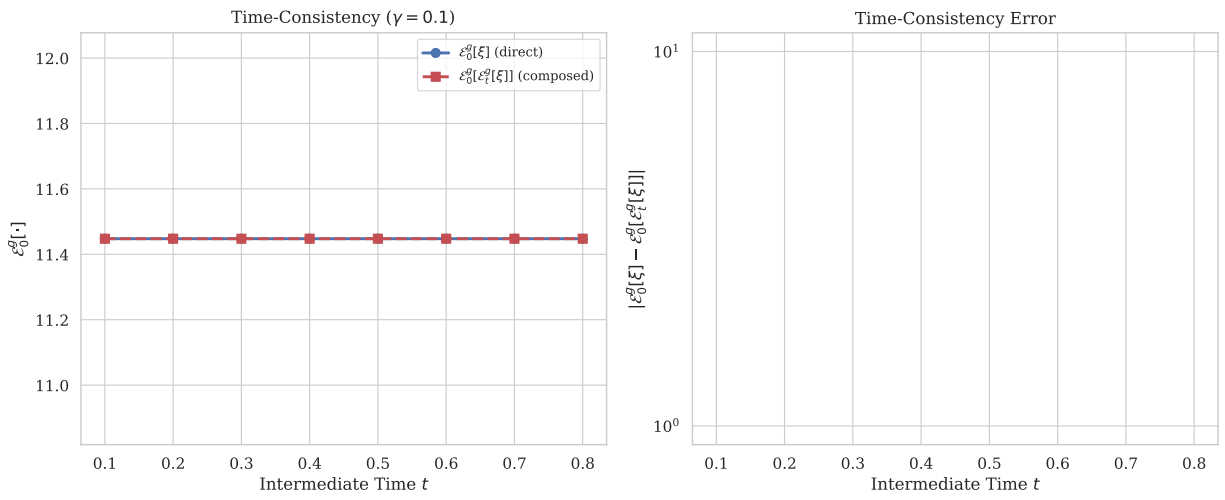


Figure 16: Time-consistency verification for $\gamma = 0.1$. Left: direct vs composed g -expectation at $t = 0$ for varying intermediate times. Right: absolute error between the two computations.